ICT702 Data Visualisation

Introducing Data Visualisation

Module 1

USC

I would like to acknowledge the traditional custodians of the land we are meeting on,

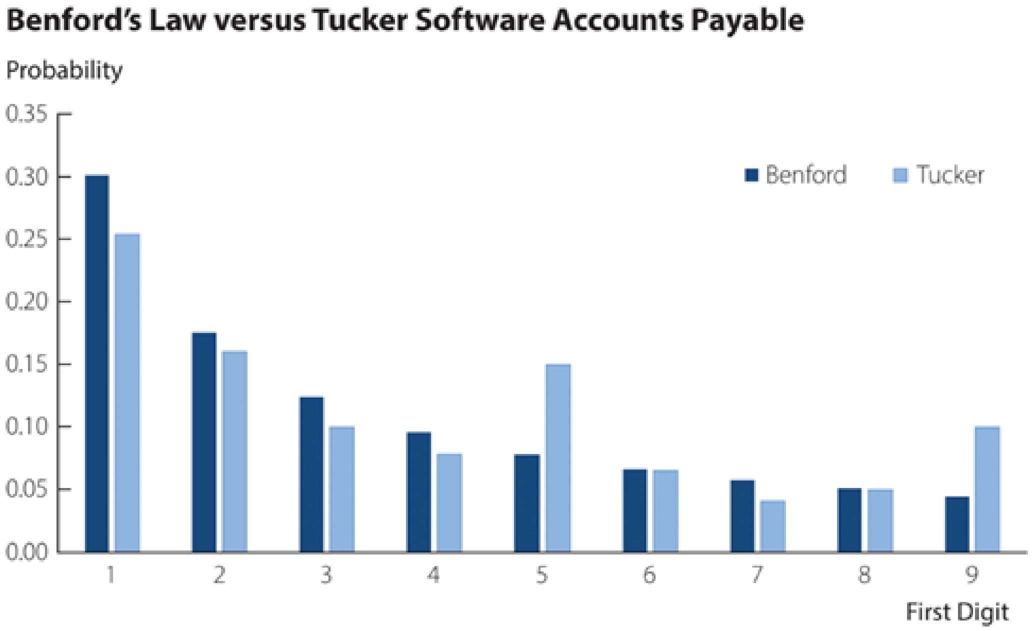
The Gubbi Gubbi/Kabi Kabi peoples,

and pay my respect to Elders past, present and emerging.

Part 2

Data Visualisation for Accounting

A clustered column chart showing Benford's Law versus Tucker Software's Accounts Payable Entries

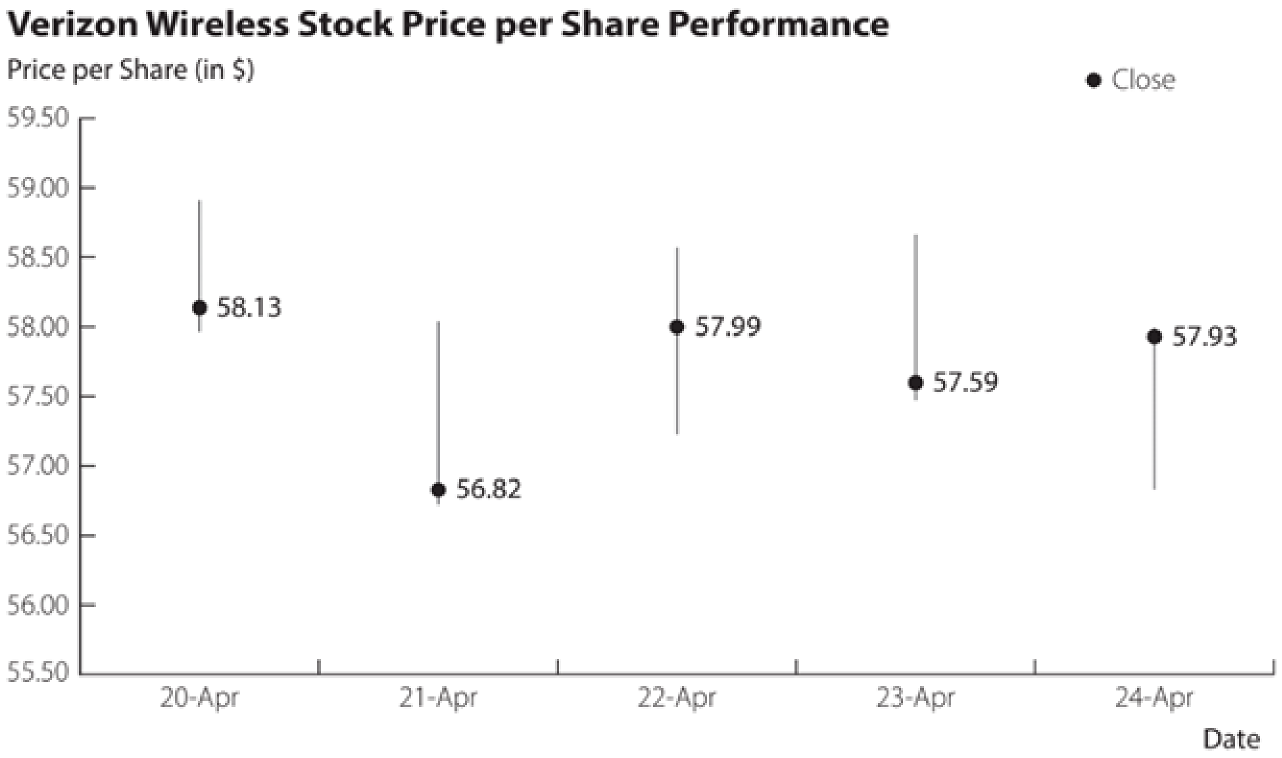


Benford's Law, (the First-Digit Law), states that the proportion of observations in which the first digit is 1 through 9, respectively, follows given probabilities.

Benford's Law may help detect fraud. If the first digits of numbers in a data set do not conform to Benford's Law, fraud investigation may be warranted.

Data Visualisation for Finance

A high-low-close stock chart for Verizon Wireless

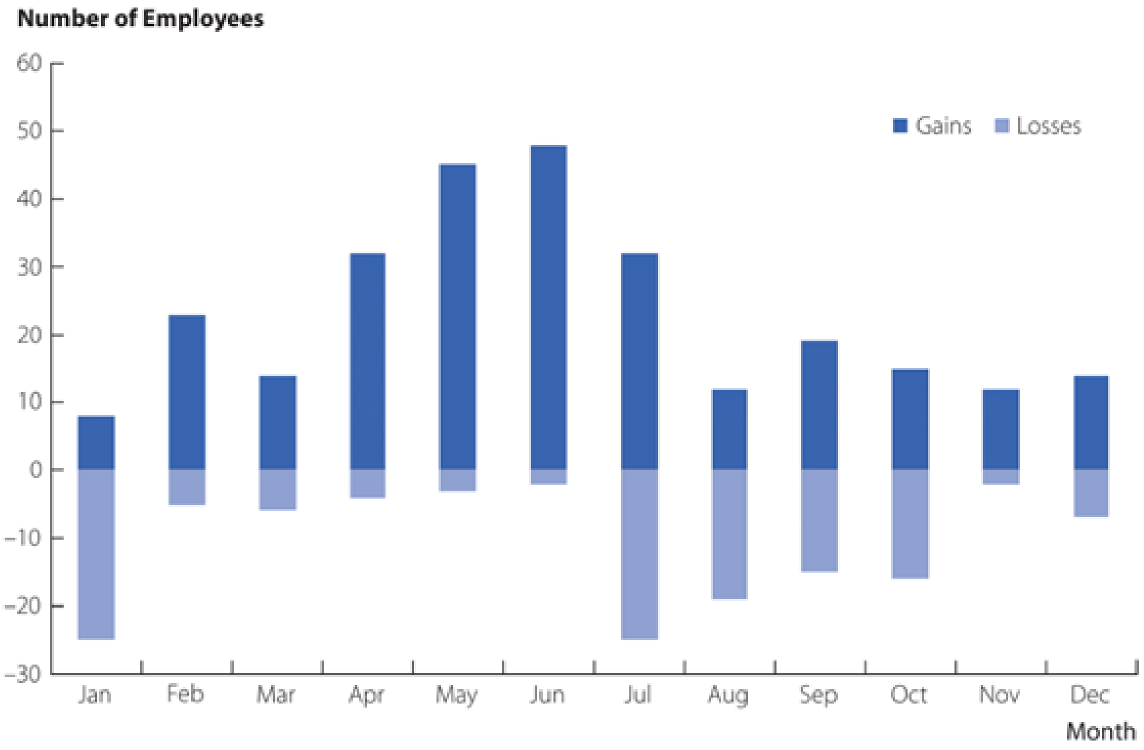


A **High-Low-Close Stock chart** shows the high value, low value, and closing value of the price of a share of stock over time.

This chart shows how the closing price is changing over time and the price volatility each day.

Data Visualisation for Human Resource Management

A stacked column chart of employee turnover by month

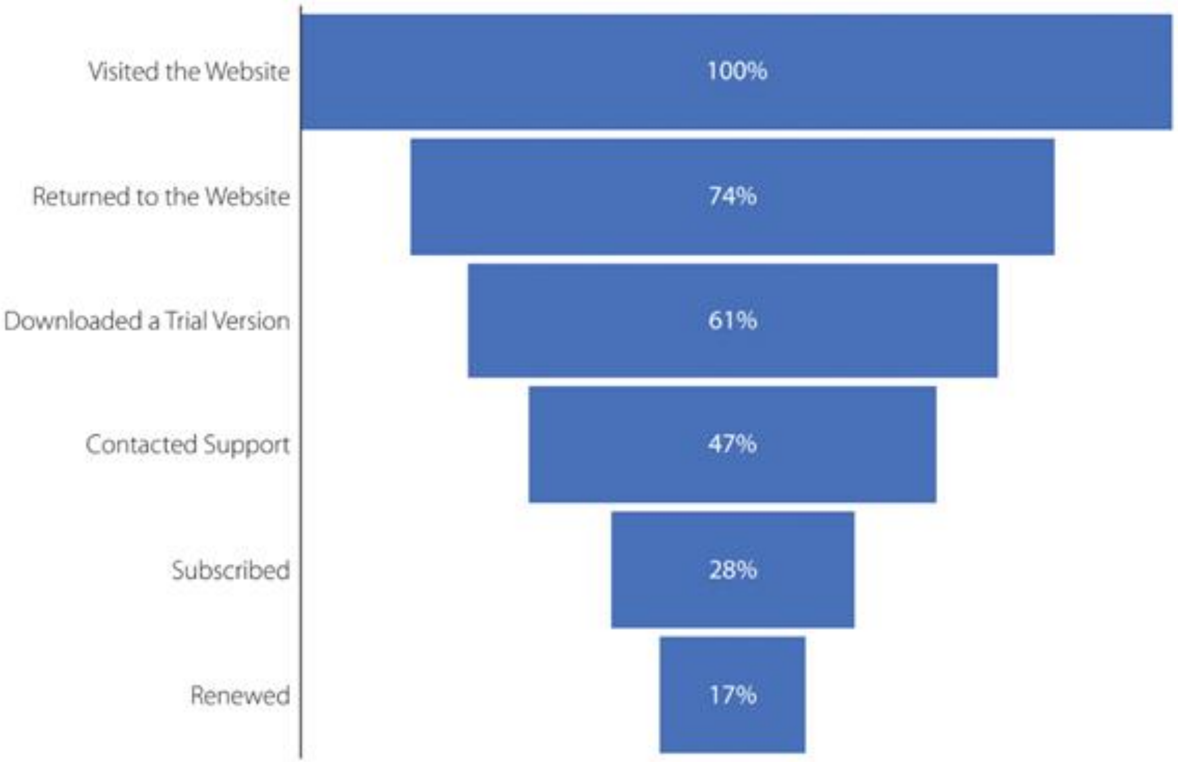


A **stacked column chart** shows part-to-whole comparisons, either over time or across categories, using different colors or shades of color.

This chart shows how January and July-October are the months in which most employees left the company, and April through June the months with most new hires.

Data Visualisation for Marketing

A funnel chart of website conversions for a software company

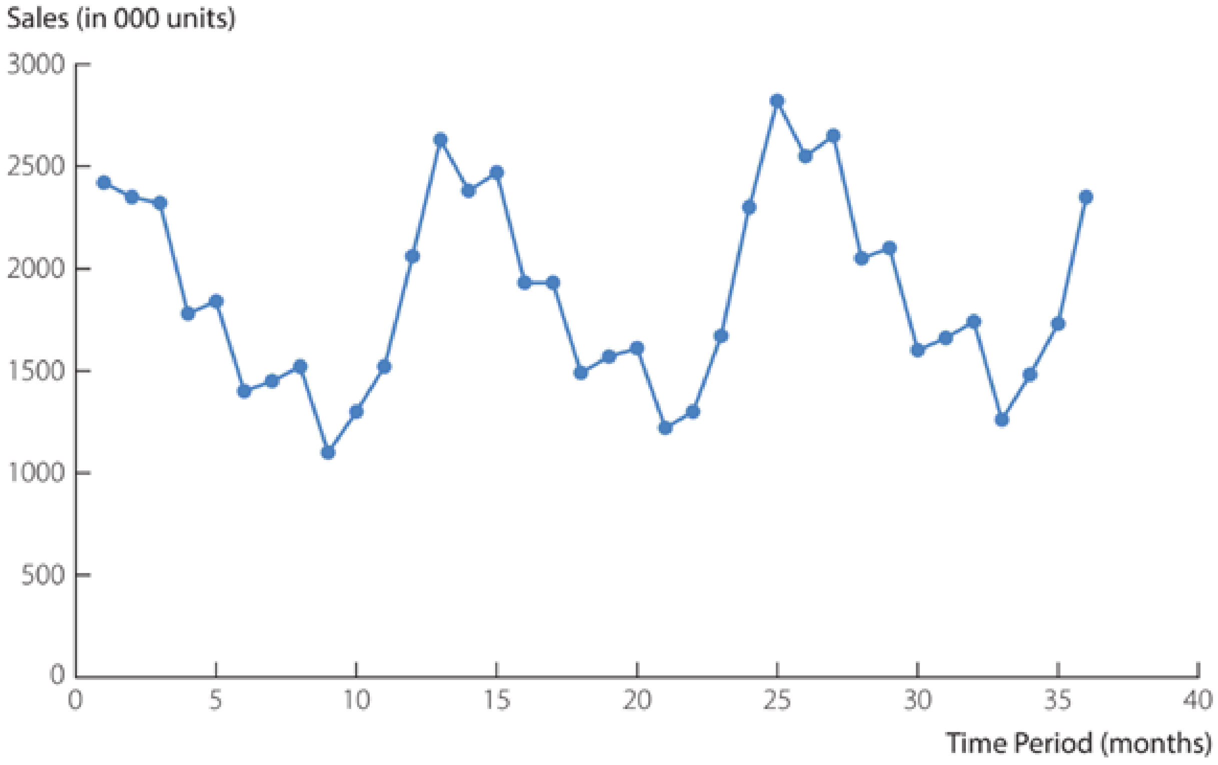


A **funnel chart** shows the progression of a numerical variable for various categories from larger to smaller values.

This chart helps to compare the conversion effectiveness of different website configurations, the use of bots, or changes in support services.

Data Visualisation for Operations

Time series data for unit sales of a product

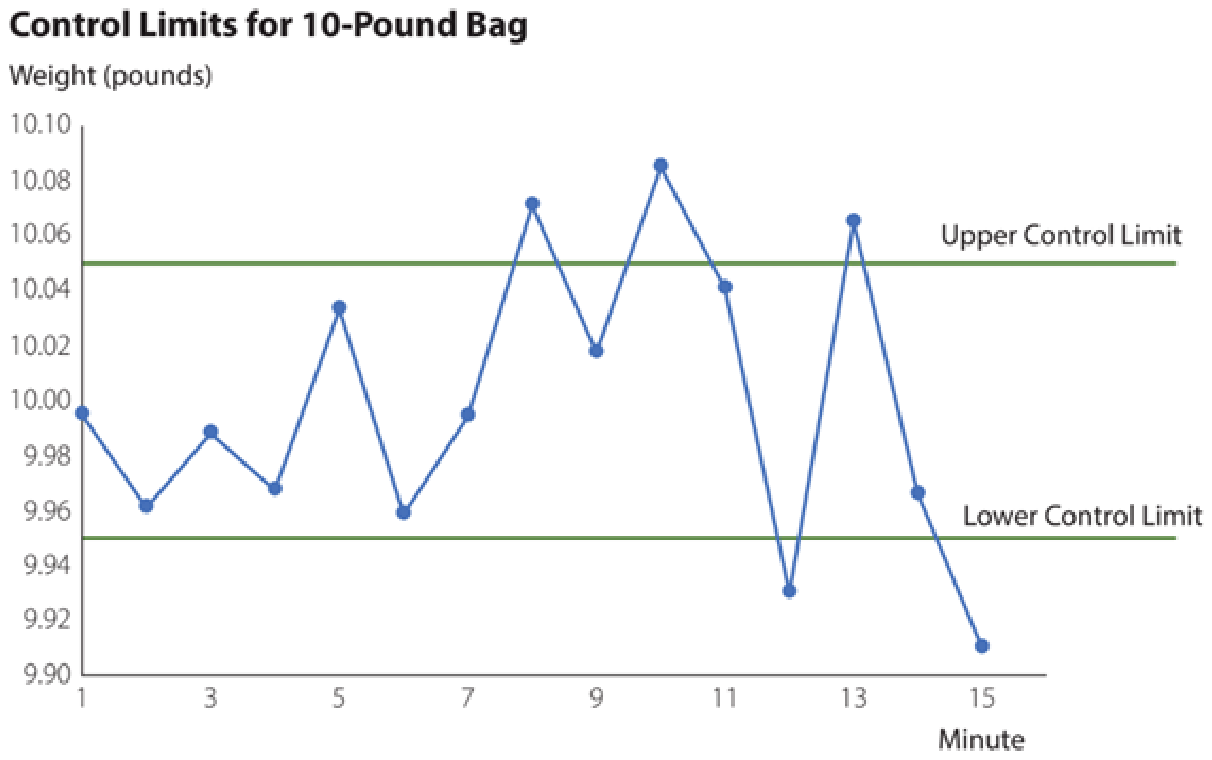


A **line chart** shows a variable of interest plotted over several time periods. A line chart helps to understand what happened in the past, identify trends over time, and project future levels.

This chart helps identify a repeating pattern and shows how units sold might also be increasing slightly over time.

Data Visualisation for Engineering

A quality control chart for dog food production



A **control chart** is a graphical display of a variable of interest plotted over time relative to lower and upper control limits. It helps determine if a production process is in or out of control.

Points beginning to appear outside the control limits are signals to inspect the process and make any necessary corrections.

Data Visualisation for Science

A spaghetti chart of hurricane tracks from multiple predictive models

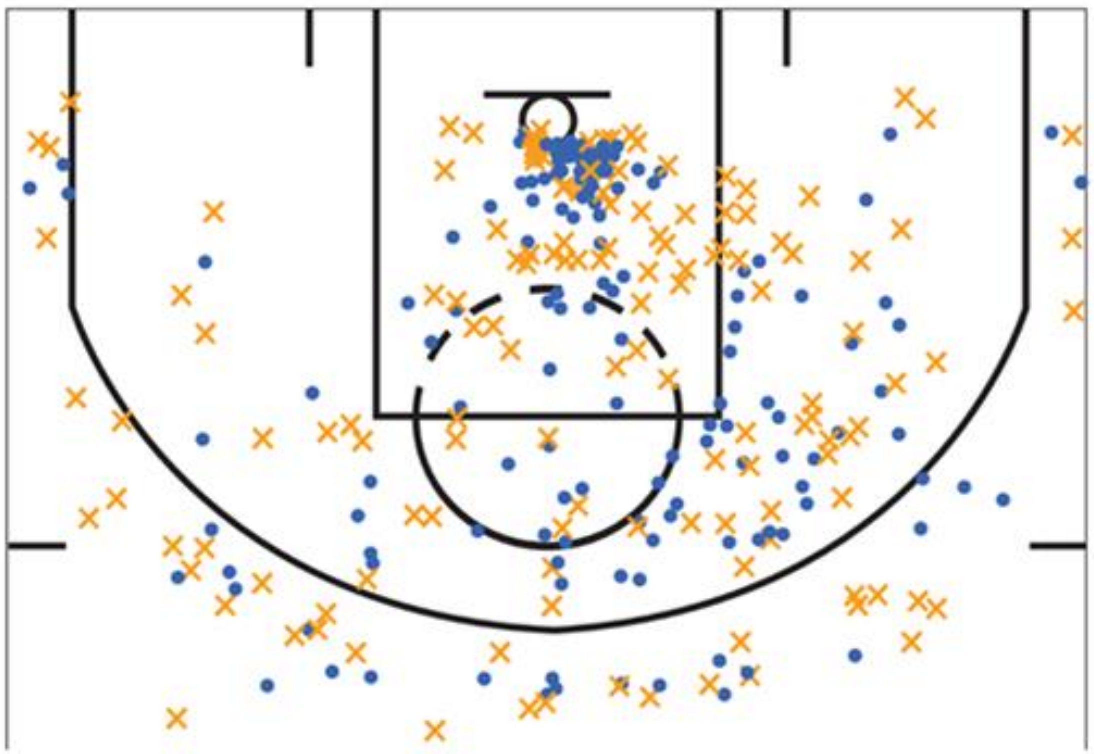
Geographic maps help display the results of complicated predictive models, such as predicting the path of a hurricane.

A **spaghetti chart** owes its name to the fact that the depiction of multiple flows through a system using a line for each possible path resemble spaghetti

Source: clickorlando.com

Data Visualisation for Sports

A shot chart for NBA player Chris Paul



A **shot chart** displays the location of the shots attempted by a player during a basketball game with different symbols or colors indicating the outcome of a shot.

This chart shows shot attempts by NBA player Chris Paul, with a blue dot indicating a successful shot and an orange x a missed shot.

Source: basketball-reference.com

Summary

In this module you have learned:

- to define analytics,
- why analytics has seen explosive growth recently,
- that data visualisation is part of descriptive analytics,
- to distinguish between major data types,
- to plan for data visualisation challenges of big data, and
- to use data visualisation techniques in major functional areas.

End of Part 2